

Class 6 Homework

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Protein B-factor Analysis Function

Q6. How would you generalize the original code above to work with any set of input protein structures? Write your own function starting from the code above that analyzes protein drug interactions by reading in any protein PDB data and outputs a plot for the specified protein.

Explanation: This function generalizes the original repeated code so it can analyze any protein structure from the PDB. The function takes a PDB ID as input, reads the structure, trims it to a specified chain and atom type, extracts B-factor values, and generates a plot. This approach improves efficiency by removing duplicated code and making the analysis reusable for multiple proteins.

```
library(bio3d)

## Function to analyze protein B-factors
plot_pdb_bfactor <- function(pdb_id, chain = "A", elety = "CA") {

  ## INPUTS:
  ## pdb_id: character string of the PDB ID (e.g., "4AKE")
  ## chain: chain identifier (default = "A")
  ## elety: atom type (default = "CA")

  ## Read PDB structure
  pdb <- read.pdb(pdb_id)

  ## Trim to selected chain and atom type
  pdb_trim <- trim.pdb(pdb, chain = chain, elety = elety)

  ## Extract B-factor values
  bvals <- pdb_trim$atom$b

  ## Plot B-factors
```

```

plotb3(bvals, sse = pdb_trim, typ = "l",
       ylab = "B-factor",
       main = paste("B-factor plot for", pdb_id))

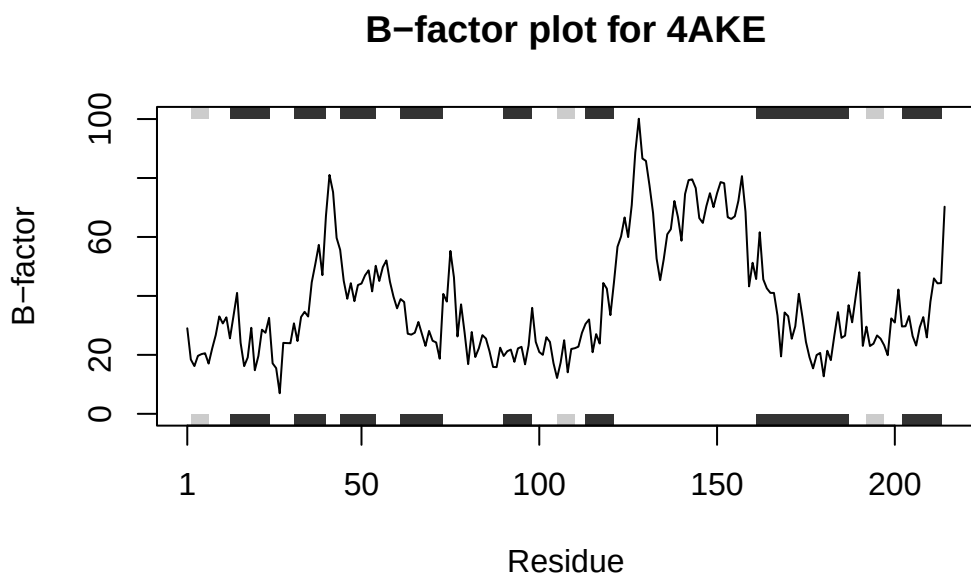
## OUTPUT:
## Returns a numeric vector of B-factor values
return(bvals)
}

```

Example Use:

```
plot_pdb_bfactor("4AKE")
```

Note: Accessing on-line PDB file



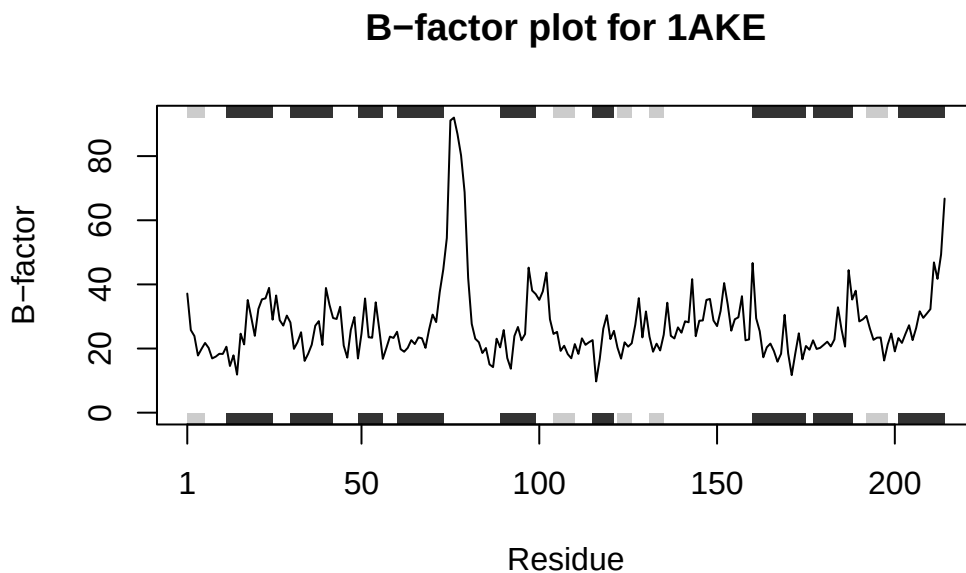
[1]	29.02	18.44	16.20	19.67	20.26	20.55	17.05	22.13	26.71	33.05
[11]	30.66	32.73	25.61	33.19	41.03	24.09	16.18	19.14	29.19	14.79
[21]	19.63	28.54	27.49	32.56	17.13	15.50	6.98	24.07	24.00	23.94
[31]	30.70	24.70	32.84	34.60	33.01	44.60	50.74	57.32	47.04	67.13
[41]	81.04	75.20	59.68	55.63	45.12	39.04	44.31	38.21	43.70	44.19
[51]	47.00	48.67	41.54	50.22	45.07	49.77	52.04	44.82	39.75	35.79

[61]	38.92	37.93	27.18	26.86	27.53	31.16	27.08	23.03	28.12	24.78
[71]	24.22	18.69	40.67	38.08	55.26	46.29	26.25	37.14	27.50	16.86
[81]	27.76	19.27	22.22	26.70	25.52	21.22	15.90	15.84	22.44	19.61
[91]	21.23	21.79	17.64	22.19	22.73	16.80	23.25	35.95	24.42	20.96
[101]	20.00	25.99	24.39	17.19	12.16	17.35	24.97	14.08	22.01	22.26
[111]	22.78	27.47	30.49	32.02	20.90	27.03	23.84	44.37	42.47	33.48
[121]	44.56	56.67	60.18	66.62	59.95	70.81	88.63	100.11	86.60	85.80
[131]	77.48	68.13	52.66	45.34	52.43	60.90	62.64	72.19	66.75	58.73
[141]	74.57	79.29	79.53	76.58	66.40	64.76	70.48	74.84	70.11	74.82
[151]	78.61	78.24	66.70	66.10	67.01	72.28	80.64	68.54	43.23	51.24
[161]	45.72	61.60	45.61	42.57	41.03	41.02	33.34	19.48	34.38	33.11
[171]	25.48	29.68	40.71	32.91	24.41	19.20	15.43	19.93	20.66	12.72
[181]	21.40	18.21	26.68	34.50	25.77	26.52	36.85	31.05	39.84	48.03
[191]	23.04	29.57	23.00	23.80	26.59	25.49	23.25	19.89	32.37	30.97
[201]	42.16	29.64	29.69	33.15	26.38	23.17	29.35	32.80	25.92	38.01
[211]	45.95	44.26	44.35	70.26						

```
plot_pdb_bfactor("1AKE")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE



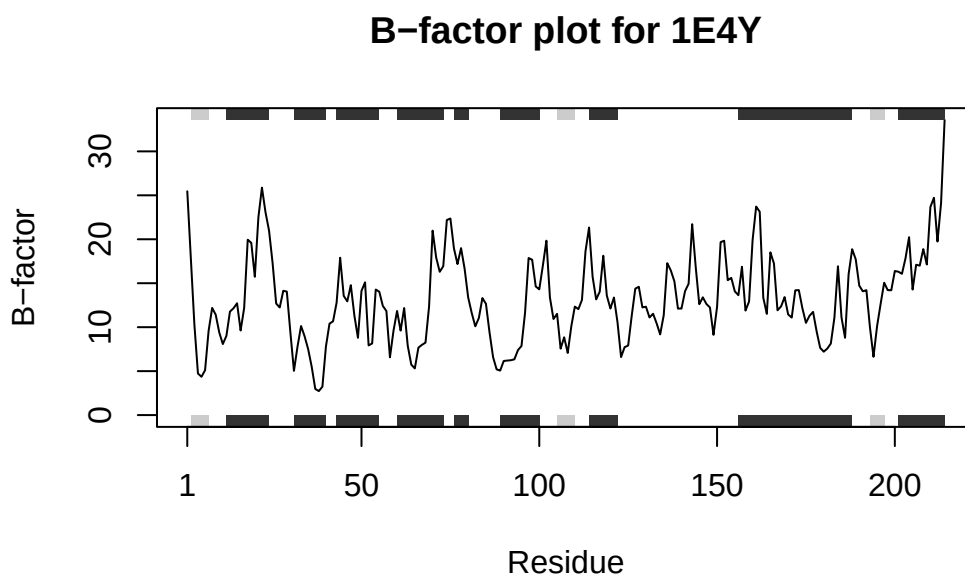
```

[1] 37.14 25.76 23.90 17.83 19.86 21.75 20.21 16.92 17.47 18.35 18.31 20.57
[13] 14.56 17.87 11.87 24.63 21.29 35.13 29.68 23.96 32.34 35.34 35.64 38.91
[25] 29.00 36.55 28.83 27.15 30.28 28.13 19.90 21.95 25.07 16.15 18.35 21.19
[37] 27.13 28.55 21.10 38.88 33.63 29.51 29.21 33.01 20.92 17.17 25.84 29.80
[49] 16.89 24.66 35.62 23.52 23.37 34.41 25.96 16.79 20.20 23.72 23.29 25.23
[61] 19.81 19.00 20.21 22.62 21.40 23.47 23.20 20.21 25.90 30.58 28.25 37.60
[73] 44.66 54.46 91.10 92.02 86.85 80.21 68.72 42.01 27.69 23.06 21.98 18.60
[85] 20.17 15.06 14.20 23.07 20.36 25.76 17.02 13.71 23.88 26.72 22.58 24.51
[97] 45.23 38.07 36.97 35.17 37.83 43.69 29.14 24.56 25.20 19.27 20.88 18.27
[109] 16.96 21.38 18.33 23.18 21.15 21.97 22.63 9.74 16.71 26.18 30.39 22.95
[121] 25.51 20.28 16.86 21.94 20.59 21.64 27.42 35.72 23.47 31.57 23.71 19.01
[133] 21.52 19.40 24.32 34.28 23.96 23.14 26.60 24.94 28.49 28.18 41.64 23.85
[145] 28.67 28.76 35.16 35.46 28.74 26.99 31.74 40.41 33.73 25.57 29.13 29.74
[157] 36.32 22.58 22.82 46.67 29.44 25.40 17.27 20.38 21.55 19.19 15.89 18.37
[169] 30.51 18.47 11.70 18.45 24.75 16.63 20.80 19.62 22.56 19.87 20.22 21.16
[181] 22.13 20.66 22.82 32.86 26.04 20.60 44.44 35.28 38.03 28.46 29.10 30.19
[193] 26.17 22.71 23.39 23.44 16.27 21.26 24.67 19.12 23.26 21.75 24.59 27.26
[205] 22.63 26.40 31.60 29.57 30.90 32.29 46.86 41.73 49.31 66.76

```

```
plot_pdb_bfactor("1E4Y")
```

Note: Accessing on-line PDB file



[1]	25.46	17.86	10.28	4.73	4.36	5.10	9.59	12.19	11.41	9.39	8.08	9.01
[13]	11.77	12.15	12.72	9.62	12.18	19.95	19.59	15.73	22.51	25.87	23.08	20.97
[25]	17.28	12.69	12.24	14.14	14.05	9.38	5.03	7.78	10.13	8.96	7.50	5.48
[37]	2.97	2.73	3.23	7.81	10.40	10.67	12.79	17.90	13.56	12.94	14.78	11.31
[49]	8.79	14.13	15.10	7.92	8.15	14.28	14.04	12.42	11.84	6.57	9.59	11.84
[61]	9.61	12.18	7.89	5.74	5.31	7.67	7.99	8.24	12.34	20.98	17.93	16.30
[73]	16.94	22.19	22.36	18.96	17.18	18.99	16.65	13.39	11.61	10.10	11.03	13.31
[85]	12.66	9.44	6.60	5.20	5.06	6.16	6.20	6.24	6.34	7.39	7.86	11.66
[97]	17.87	17.67	14.63	14.30	16.98	19.84	13.36	10.93	11.52	7.56	8.85	7.07
[109]	10.08	12.34	12.05	13.10	18.63	21.34	15.73	13.16	14.04	18.13	13.59	12.12
[121]	13.37	10.57	6.60	7.73	7.91	11.31	14.38	14.60	12.25	12.33	11.10	11.53
[133]	10.44	9.18	11.36	17.28	16.45	15.21	12.11	12.12	14.10	14.94	21.72	16.82
[145]	12.61	13.40	12.64	12.24	9.13	12.31	19.68	19.83	15.34	15.61	14.07	13.64
[157]	16.87	11.89	12.92	19.93	23.72	23.13	13.35	11.51	18.51	17.24	11.92	12.36
[169]	13.42	11.45	11.09	14.19	14.22	12.15	10.49	11.29	11.74	9.53	7.65	7.21
[181]	7.56	8.14	11.07	16.93	11.12	8.79	16.03	18.87	17.72	14.72	14.08	14.21
[193]	9.99	6.63	10.11	12.64	15.06	14.21	14.20	16.39	16.31	16.07	17.83	20.24
[205]	14.28	17.10	17.00	18.88	17.13	23.68	24.72	19.74	24.12	33.57		